AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-0841	**Classification:**	Major / Critical Quality Attribute (CQA)
Product / API Name:	Atorvastatin Calcium Trihydrate API (USP/EP Grade)	**Batch / Lot Number:**	ATC-2026-B09
Manufacturing Stage:	Stage 4: Crystallization, Filtration & Vacuum Drying	**Equipment Tag:**	Reactor CRYS-RX-04 / Filter-Dryer ANFD-02
Date of Occurrence:	2026-10-22 14:15 UTC	**Date of Discovery:**	2026-10-23 09:30 UTC
Material / Lot ID:	Ethanol Wash RAW-ETH-4029	**Reporting Unit:**	QC Analytical Testing Laboratory

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Related Substance: Impurity D	**NMT 0.15% (w/w)**	**0.28% (w/w) [OOS]**	USP HPLC (Waters Alliance #HPLC-8812)
Assay (Anhydrous Basis)	98.0% - 102.0%	99.4%	HPLC UV Detection at 246 nm
Water Content (Karl Fischer)	3.5% - 5.5%	4.1%	Mettler Toledo V30S Titrator
Residual Ethanol	NMT 5000 ppm	1820 ppm	Headspace GC-FID

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

During final release testing of Atorvastatin Calcium Trihydrate API Batch ATC-2026-B09, QC analyst reported Impurity D at 0.28% (w/w), exceeding the specification limit of <= 0.15% (w/w). Phase 1 laboratory investigation confirmed no testing fault. Inspection of ANFD-02 revealed that during vacuum drying sequence at 14:15, jacket heating control valve V-104 failed to throttle closed. Temperature spiked to 58°C and remained elevated for 45 minutes before manual operator override.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. All 12 fiberboard drums of Batch ATC-2026-B09 placed on QA Quarantine status in Warehouse 3.
2. Dryer ANFD-02 locked out in SCADA; heating valve actuator isolated for mechanical inspection.
3. Raw material solvent ethanol lot RAW-ETH-4029 quarantined.

Prepared by:	R. Patel (QC Senior Chemist)	2026-10-23
Verified by:	M. Vance (Production Manager)	2026-10-23
QA Approval:	Dr. S. Thornton (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**